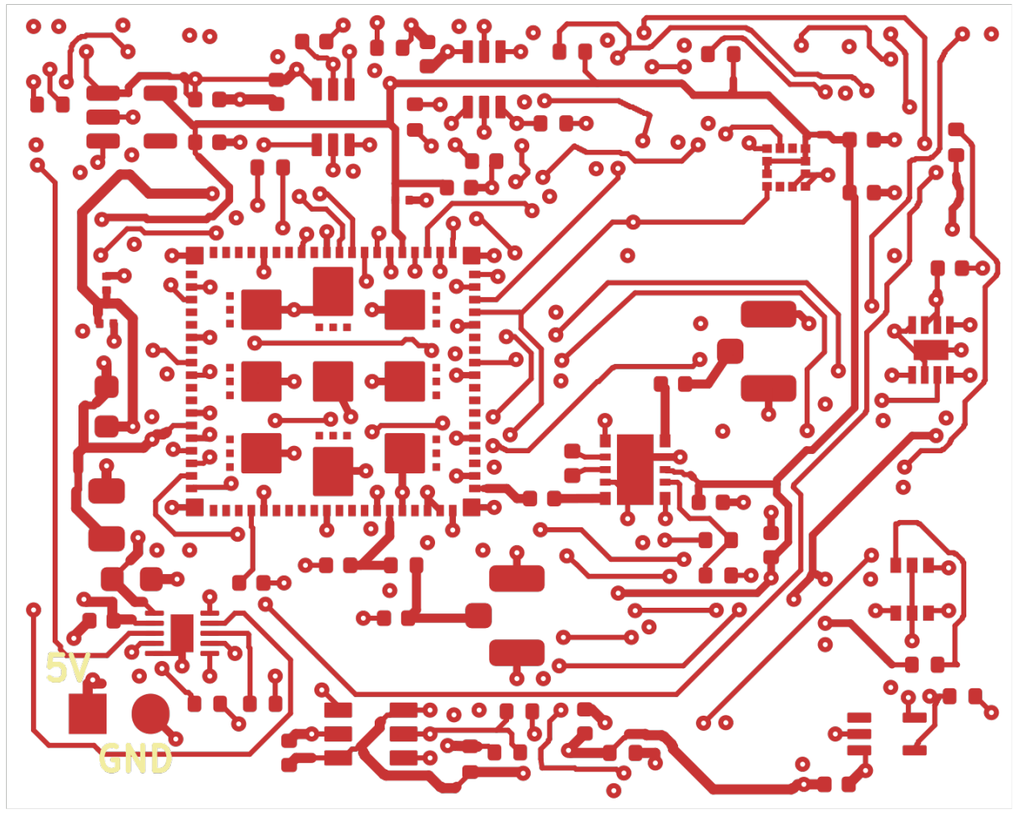
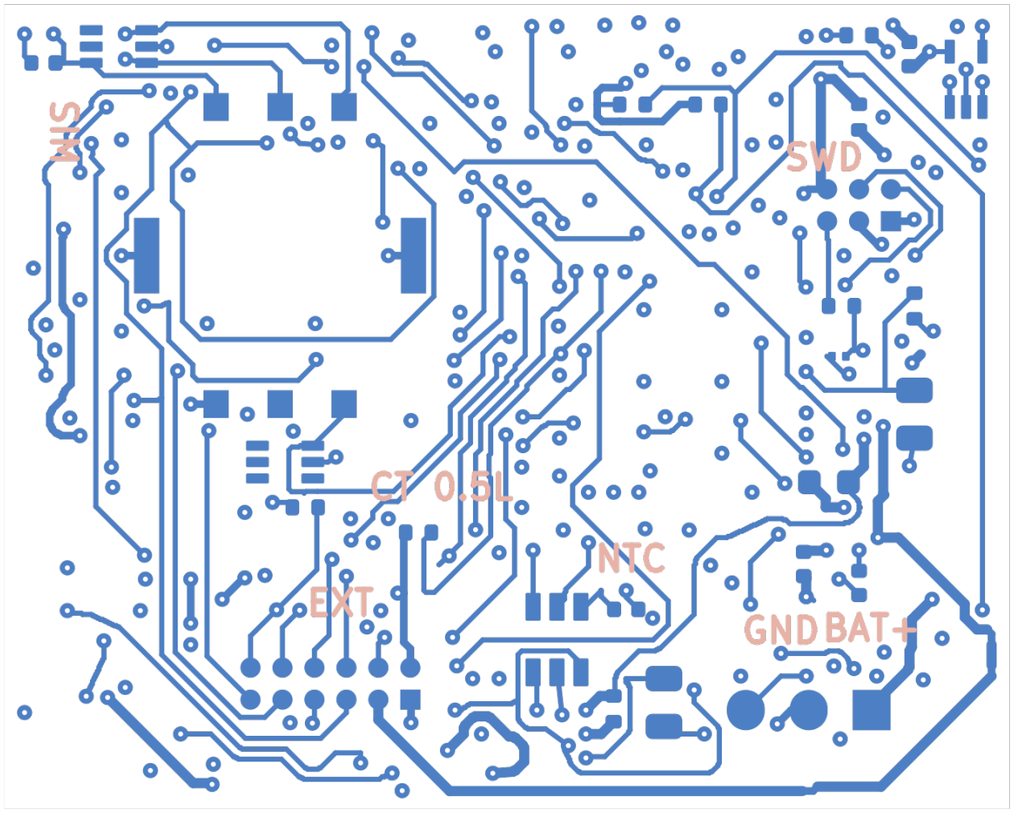
COLLAR TRACKER / VERIFIED PCB

r05l PCB DFM is complete. A fully assembled, working tracker is not released.

TOP / 40 x 32mm



BOTTOM / mirrored view



VERIFIED / UNCHANGED

Six copper layers; physical nano-SIM.

Native DRC / ERC / opens / parity: zero findings.

Both JLC PCB runs: 0 Danger / 0 Warning / 455 Good.

Same ZIP and rules; all 29 rows analyzed; SMT off.

81 fitted references / 44 candidate MPNs.

109 physical GND pads have local via paths.

Independent hardware hot/cold charge interlock.

REMAINING BEFORE A WORKING-DEVICE ORDER

Protected pack and lead-exit approval.

Dock holding and wire-end restraint.

Internal contact lugs / J3 wiring and joining process.

Factory sourcing, CPL rotation and SMT review.

Firmware, programming and functional tests.

Powered / RF / physical water and dust tests.

No purchase, supplier message or physical test made.

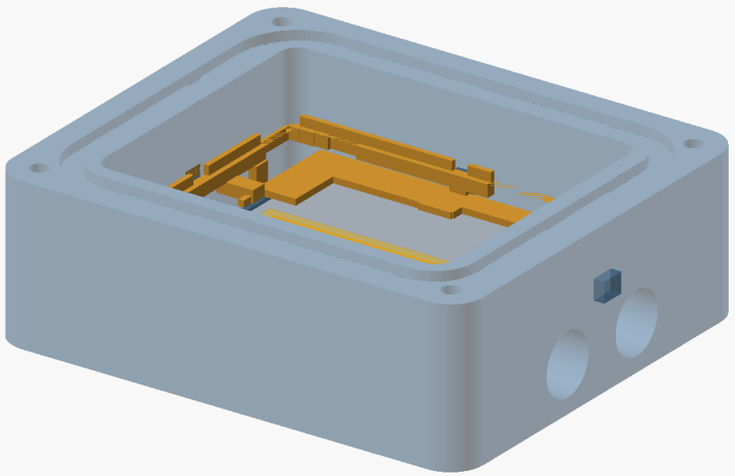
Gerber SHA256: c736c5d35ddfe4c4662e57606120f00f7358492e11d5fbb32eaf1d0ff6caf5d2

JLC records: 637366592129687553 / 637366801595977730

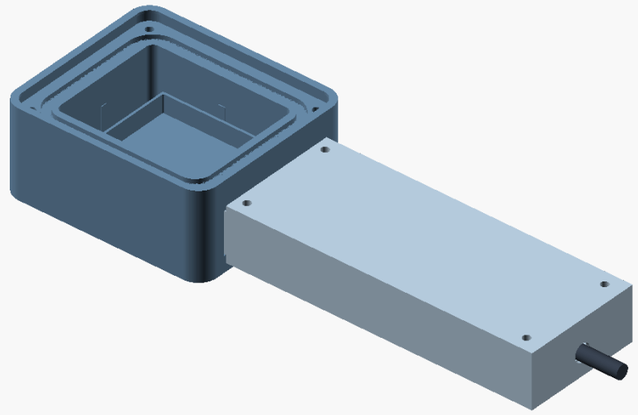
COLLAR TRACKER / CURRENT ASSEMBLY STUDY

Conditional geometry. Pack supplier response, factory assembly and physical validation remain open.

r12 / requested pack and retaining cap



r13 shell / r14 dock; lid and internals omitted



CURRENT MODEL

Rigid shell + lid: 57.4 x 49.4 x 21.0mm.

Body including external key: 61.4mm in X.

Boot: 60.4 x 52.4 x 25.8mm.

r12 carrier/cap; r11 lid, frame, boot and seals.

r13 shell/key; r14 dock: 94 x 36 x 16mm.

Constant-length pigtails; continuous radius >=8.696mm.

Full wire-family / housing CAD intersection empty.

LIMITS / DO NOT ORDER AS A FINISHED DEVICE

Requested pack: 32 x 30.5 x 5.3mm maximum.

That pack envelope is not supplier approved.

Free-wire geometry does not constrain real buckling.

Contact holding / clamp / cycle-life tests remain open.

Charging is a dry indoor prototype only.

IP68 is a target, not a certified or measured result.

2m fresh water / 60min is an engineering assumption.

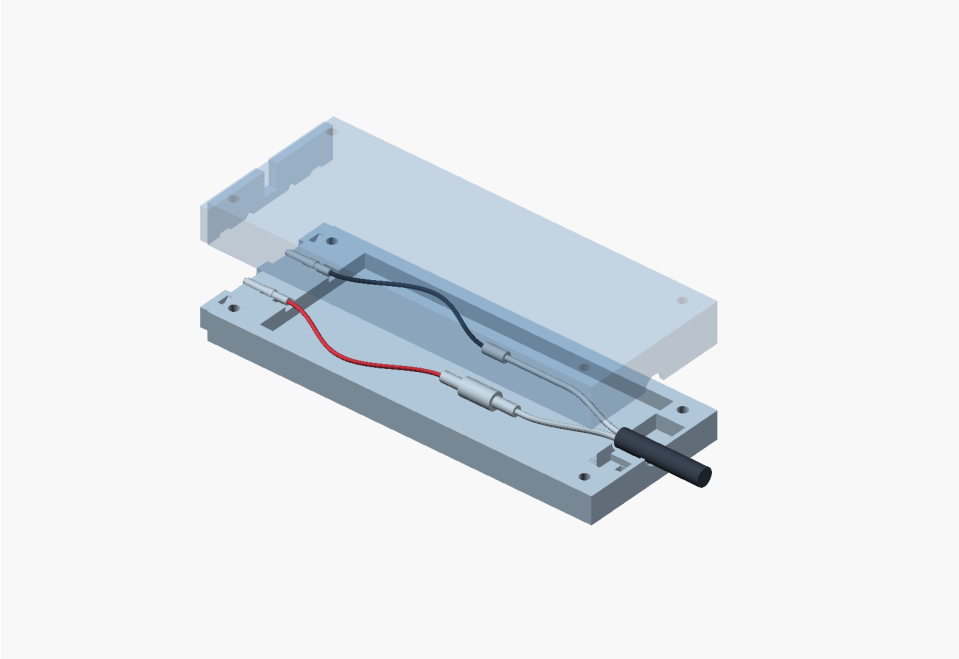
Parts in this review are explicitly revision-mapped; previous releases remain preserved.

Open the repository STATUS / RESUME for the exact source, commands and remaining work.

CT-DOCK-001-B | Charging-wire movement / r14

GEOMETRY STUDY — fixed-end restraint, contact holding and physical qualification remain open

External dock: 94 x 36 x 16mm

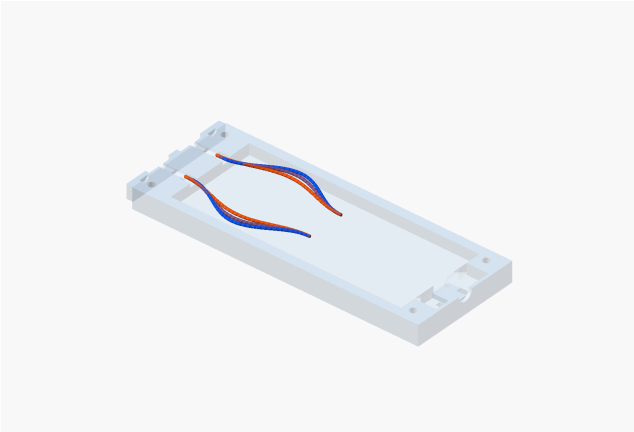


Wearable, PCB, seals, key and contact seats are unchanged.

USB red → fuse → Y=-6 → J3.1 /VIN_5V

USB black → Y=+6 → J3.2 /GND

Constant-length wire family



Free span at 23C	40.95 +/- 0.10mm per wire
In-service span bound	40.775 to 41.125mm
Conservative contact travel	2.1421mm
Minimum remaining slack	0.3858mm
Continuous radius bound	8.6961mm (required 8.382mm)
Full sweep / housing	Empty CAD intersection
Old housing control	Interference detected

The complete calculated family fits. Three colored poses illustrate it; interval bounds establish the continuous envelope.

This does not constrain a free wire to that shape. A fixed-end clamp, forming process and cyclic-flex tests are still required.

40.95mm excludes stripped conductor, crimp insertion and splice overlap. Do not use it as a production cut length.

The r05I PCB retains both recorded 0 Danger / 0 Warning / 455 Good JLC runs. No unchanged-byte DFM was repeated.

Dock retention, internal J3 wiring, pack approval, firmware and physical powered/RF/ingress tests remain open.

This dry indoor charging prototype is not an order release. IP68 is a target, not a measured or certified result.